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Artificial muscles comprising a partially insulated electrode pair and methods for optimizing voltage polarity conditions in the artificial muscles

Abstract

An artificial muscle system includes an artificial muscle and a power supply. The artificial muscle includes an electrode pair including a first electrode and a second electrode, an electrical insulator membrane fixed to the second electrode, a housing including an electrode region and an expandable fluid region, the electrode pair positioned in the electrode region of the housing, and a dielectric fluid housed within the housing. The power supply includes a positive terminal and a negative terminal. The positive terminal is electrically coupled to the second electrode. The negative terminal is electrically coupled to the first electrode. The electrode pair is actuatable between a non-actuated state and an actuated state such that actuation from the non-actuated state to the actuated state directs the dielectric fluid into the expandable fluid region.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2020/0032822	12/2019	Keplinger	N/A	F15B 21/06
2020/0132213	12/2019	Gandhi	N/A	H02N 1/006
2021/0369547	12/2020	Mau et al.	N/A	N/A
2021/0370499	12/2020	Rowe	N/A	F15B 21/065

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
5144096	12/2012	JP	N/A
20080028148	12/2007	KR	N/A
20080028171	12/2007	KR	N/A
20210086518	12/2020	KR	N/A
2008035732	12/2007	WO	N/A

OTHER PUBLICATIONS

Kyung et al, Pouch Type Soft Actuator with Hetero Space Change Layer and Dielectric Fluid, Jul. 8, 2021, KR 20210086518 (English Machine Translation) (Year: 2021). cited by examiner
E. Acome et al. "Hydraulically amplified self-healing electrostatic actuators with muscle-like performance", Science Jan. 5, 2018: vol. 359, Issue 6371, pp. 61-65. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the priority benefit of U.S. Provisional Patent Application No. 63/285,707, filed Dec. 3, 2021, for “Artificial Muscles Comprising A Partially Insulated Electrode Pair And Methods For Optimizing Voltage Polarity Conditions In The Artificial Muscles,” which is hereby incorporated by reference in its entirety including the drawings.

TECHNICAL FIELD

(1) The present specification generally relates to apparatus and methods for focused inflation on at least one surface of a device, and, more specifically, apparatus and methods for utilizing an electrode pair to direct a fluid to inflate the device.

BACKGROUND

(2) Current robotic technologies rely on rigid components, such as servomotors to perform tasks, often in a structured environment. This rigidity presents limitations in many robotic applications, caused, at least in part, by the weight-to-power ratio of servomotors and other rigid robotics devices. The field of soft robotics improves on these limitations by using artificial muscles and other soft actuators. Artificial muscles attempt to mimic the versatility, performance, and reliability of a biological muscle. Some artificial muscles rely on fluidic actuators, but fluidic actuators require a supply of pressurized gas or liquid, and fluid transport must occur through systems of channels and tubes, limiting the speed and efficiency of the artificial muscles. Other artificial muscles use thermally activated polymer fibers, but these are difficult to control and operate at low efficiencies. Additionally, electrically coupling a non-insulated electrode to a positive terminal of a power supply may result in a chemical buildup, e.g., oligomerization, on the exposed surface of the non-insulated electrode in contact with a dielectric fluid. This chemical buildup may result in operation failure of the artificial muscle.

(3) Accordingly, a need exists for improved artificial muscles that are electrically coupled to a power supply in a particular manner based on whether an electrical insulator layer is provided on one or both of the electrodes of the artificial muscles.

SUMMARY

(4) In one embodiment, an artificial muscle system includes an artificial muscle and a power supply. The artificial muscle includes an electrode pair including a first electrode and a second electrode, an electrical insulator membrane fixed to a surface of the second electrode facing the first electrode, a housing including an electrode region and an expandable fluid region, the electrode pair positioned in the electrode region of the housing, the first electrode fixed to a first surface of the housing and the second electrode fixed to a second surface of the housing, and a dielectric fluid housed within the housing. The power supply includes a positive terminal and a negative terminal. The positive terminal is electrically coupled to the second electrode. The negative terminal is electrically coupled to the first electrode. The electrode pair is actuatable between a non-actuated state and an actuated state such that actuation from the non-actuated state to the actuated state directs the dielectric fluid into the expandable fluid region.

(5) In another embodiment, an artificial muscle system including an artificial muscle and a power supply. The artificial muscle includes an electrode pair including a first electrode and a second electrode, the first electrode and the second electrode each including a plurality of electrode segments, an electrical insulator membrane fixed to one or more electrode segments of the first electrode and one or more corresponding electrode segments of the second electrode, a housing comprising an electrode region and an expandable fluid region, the electrode pair positioned in the electrode region of the housing, the first electrode fixed to a first surface of the housing and the second electrode fixed to a second surface of the housing, and a dielectric fluid housed within the housing. The power supply includes a positive terminal and a negative terminal, the positive terminal electrically coupled to the one or more electrode segments of the first electrode and the one or more corresponding electrode segments of the second electrode, the negative terminal

electrically coupled to one or more other electrode segments of the first electrode and one or more other electrode segments of the second electrode. The electrode pair is actuatable between a non-actuated state and an actuated state such that actuation from the non-actuated state to the actuated state directs the dielectric fluid into the expandable fluid region.

(6) In yet another embodiment, a method for actuating an artificial muscle includes electrically coupling a negative terminal and a positive terminal of a power supply to an electrode pair of the artificial muscle. The artificial muscle includes a housing having an electrode region and an expandable fluid region. The electrode pair includes a first electrode electrically coupled to the negative terminal of the power supply, and a second electrode electrically coupled to the positive terminal of the power supply, the electrode pair positioned in the electrode region of the housing. An electrical insulator membrane is fixed to the second electrode. The method further includes applying a voltage from the power supply to the electrode pair of the artificial muscle, thereby actuating the electrode pair from a non-actuated state to an actuated state such that dielectric fluid is directed into the expandable fluid region of the housing and expands the expandable fluid region.

(7) These and additional features provided by the embodiments described herein will be more fully understood in view of the following detailed description, in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The embodiments set forth in the drawings are illustrative and exemplary in nature and not intended to limit the subject matter defined by the claims. The following detailed description of the illustrative embodiments can be understood when read in conjunction with the following drawings, where like structure is indicated with like reference numerals and in which:

(2) FIG. 1 schematically depicts an exploded view of an artificial muscle, according to one or more embodiments shown and described herein;

(3) FIG. 2 schematically depicts a top view of the artificial muscle of FIG. 1, according to one or more embodiments shown and described herein;

(4) FIG. 3 schematically depicts a cross-sectional view of the artificial muscle of FIG. 1 taken along line 3-3 in FIG. 2 in a non-actuated state, according to one or more embodiments shown and described herein;

(5) FIG. 4 schematically depicts a cross-sectional view of the artificial muscle of FIG. 3 in an actuated state, according to one or more embodiments shown and described herein;

(6) FIG. 5 schematically depicts an enlarged cross-sectional view of the artificial muscle of FIG. 1 taken along line 5-5 in FIG. 3, according to one or more embodiments shown and described herein;

(7) FIG. 6 schematically depicts an enlarged cross-sectional view of an artificial muscle, according to one or more embodiments shown and described herein;

(8) FIG. 7 schematically depicts a partial plan view of an artificial muscle, according to one or more embodiments shown and described herein;

(9) FIG. 8 schematically depicts a cross-sectional view of the artificial muscle of FIG. 7 taken along line 8-8 in FIG. 7, according to one or more embodiments shown and described herein; and

(10) FIG. 9 schematically depicts an actuation system for operating an artificial muscle, according to one or more embodiments shown and described herein.

DETAILED DESCRIPTION

(11) Embodiments described herein are directed to artificial muscles and artificial muscle assemblies electrically coupled to a power supply. The artificial muscles described herein are actuatable to selectively raise and lower a region of the artificial muscles to provide a selective, on demand inflated expandable fluid region. The artificial muscles include a housing and an electrode pair. A dielectric fluid is housed within the housing, and the housing includes an electrode region

and an expandable fluid region, where the electrode pair is positioned in the electrode region. The electrode pair includes a first electrode and a second electrode. The electrode pair is actuatable between a non-actuated state and an actuated state such that actuation from the non-actuated state to the actuated state directs the dielectric fluid into the expandable fluid region. This expands the expandable fluid region, raising a portion of the artificial muscle on demand. Various embodiments of the artificial muscles and the operation of the artificial muscles are described in more detail herein. Whenever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts.

(12) Referring now to FIGS. **1** and **2**, an artificial muscle **100** is shown. The artificial muscle **100** includes a housing **102**, an electrode pair **104**, including a first electrode **106** and a second electrode **108**, fixed to opposite surfaces of the housing **102**, and an electrical insulator membrane **110** fixed to the second electrode **108**. It should be appreciated that the first electrode **106** remains uninsulated. However, in embodiments, a second electrical insulator membrane may be fixed to the first electrode **106** to electrically insulate the first electrode **106**.

(13) As discussed in more detail herein, the second electrode **108**, which receives the electrical insulator membrane **110**, is to be electrically coupled or electrically connected to a positive terminal of a power supply, such as a positive terminal **406A** of a power supply **406** (FIG. **9**). Accordingly, the first electrode **106**, which does not receive the electrical insulator membrane **110**, is to be electrically coupled or electrically connected to a negative terminal of a power supply, such as a negative terminal **406B** of the power supply **406** (FIG. **9**). Although the electrical insulator membrane **110** is described herein as being fixed to the second electrode **108**, it should be appreciated that, in embodiments, the electrical insulator membrane **110** may alternatively be fixed to the first electrode **106**. In this embodiment, the first electrode **106** would be electrically coupled to the positive terminal of the power supply, and the second electrode **108** would be electrically coupled to the negative terminal of the power supply.

(14) As referred to herein, reference to a “negative terminal” of a power supply refers to the negatively charged electrode of a power supply from which the electrons emerge, which may commonly be referred to as a “cathode.” Similarly, as referred to herein, reference to a “positive terminal” of a power supply refers to the positively charged electrode of a power supply that receives electrons, which may commonly be referred to as a “anode.”

(15) In some embodiments, the housing **102** is a one-piece monolithic layer including a pair of opposite inner surfaces, such as a first inner surface **114** and a second inner surface **116**, and a pair of opposite outer surfaces, such as a first outer surface **118** and a second outer surface **120**. In some embodiments, the first inner surface **114** and the second inner surface **116** of the housing **102** are heat-sealable. In other embodiments, the housing **102** may be a pair of individually fabricated film layers, such as a first film layer **122** and a second film layer **124**. Thus, the first film layer **122** includes the first inner surface **114** and the first outer surface **118**, and the second film layer **124** includes the second inner surface **116** and the second outer surface **120**.

(16) Throughout the ensuing description, reference may be made to the housing **102** including the first film layer **122** and the second film layer **124**, as opposed to the one-piece housing. It should be understood that either arrangement is contemplated. In some embodiments, the first film layer **122** and the second film layer **124** generally include the same structure and composition. For example, in some embodiments, the first film layer **122** and the second film layer **124** each comprises biaxially oriented polypropylene (BOPP).

(17) The first electrode **106** and the second electrode **108** are each positioned between the first film layer **122** and the second film layer **124**. In some embodiments, the first electrode **106** and the second electrode **108** are each aluminum-coated polyester such as, for example, Mylar®. In addition, one of the first electrode **106** and the second electrode **108** is a negatively charged electrode and the other of the first electrode **106** and the second electrode **108** is a positively charged electrode. For purposes discussed herein, either electrode **106**, **108** may be positively

charged so long as the other electrode **106**, **108** of the artificial muscle **100** is negatively charged.

(18) The first electrode **106** has a film-facing surface **126** and an opposite inner surface **128**. The first electrode **106** is positioned against the first film layer **122**, specifically, the first inner surface **114** of the first film layer **122**. In addition, the first electrode **106** includes a first terminal **130** extending from the first electrode **106** past an edge of the first film layer **122** such that the first terminal **130** can be connected to a power supply to actuate the first electrode **106**. Specifically, the terminal **130** is coupled, either directly or in series, to a power supply and a controller of an actuation system **400**, as shown in FIG. **9**. Similarly, the second electrode **108** has a film-facing surface **148** and an opposite inner surface **150**. The second electrode **108** is positioned against the second film layer **124**, specifically, the second inner surface **116** of the second film layer **124**. The second electrode **108** includes a second terminal **152** extending from the second electrode **108** past an edge of the second film layer **124** such that the second terminal **152** can be connected to a power supply and a controller of the actuation system **400** to actuate the second electrode **108**.

(19) It should be appreciated that the second terminal **152**, which extends from the second electrode **108** that is electrically insulated by the electrical insulator membrane **110**, is electrically coupled to the positive terminal **406A** of the power supply **406** (FIG. **9**). Additionally, the first terminal **130**, which extends from the first electrode **106** that is not electrically insulated (non-insulated), is electrically coupled to a negative terminal **406B** of the power supply **406** (FIG. **9**).

(20) By electrically coupling the insulated electrode, i.e., the second electrode **108**, to the positive terminal **406A** of the power supply **406**, a chemical buildup, e.g., oligomerization, on the non-insulated electrode, i.e., the first electrode **106**, is reduced. It should be appreciated that in instances in which a non-insulated electrode is connected to a positive terminal of a power supply, the dielectric fluid, such as dielectric fluid **198**, will build up on the non-insulated electrode and create a “scum” layer on the exposed, non-insulated surface of the electrode.

(21) In the present embodiment, the positive terminal **406A** of the power supply **406** is electrically coupled to the second electrode **108**. However, it should be appreciated that, in other embodiments, the first electrode **106** may be insulated by the electrical insulator membrane **110**, thereby leaving the second electrode **108** non-insulated. In this embodiment, the first electrode **106** would be electrically coupled to the positive terminal **406A** of the power supply **406** and the second electrode **108** would be electrically coupled to the negative terminal **406B** of the power supply **406**.

(22) With respect now to the first electrode **106**, the first electrode **106** includes two or more fan portions **132** extending radially from a center axis C of the artificial muscle **100**. In some embodiments, the first electrode **106** includes only two fan portions **132** positioned on opposite sides or ends of the first electrode **106**. In some embodiments, the first electrode **106** includes more than two fan portions **132**, such as three, four, or five fan portions **132**. In embodiments in which the first electrode **106** includes an even number of fan portions **132**, the fan portions **132** may be arranged in two or more pairs of fan portions **132**. As shown in FIG. **1**, the first electrode **106** includes four fan portions **132**. In this embodiment, the four fan portions **132** are arranged in two pairs of fan portions **132**, where the two individual fan portions **132** of each pair are diametrically opposed to one another.

(23) Each fan portion **132** has a first side edge **132a** and an opposite second side edge **132b**. As shown, the first terminal **130** extends from the second end **136** of one of the fan portions **132** and is integrally formed therewith. A channel **133** is at least partially defined by opposing side edges **132a**, **132b** of adjacent fan portions **132** and, thus, extends radially toward the center axis C. The channel **133** terminates at an end **140a** of a bridge portion **140** interconnecting adjacent fan portions **132**.

(24) As shown in FIG. **1**, dividing lines D are included to depict the boundary between the fan portions **132** and the bridge portions **140**. The dividing lines D extend from the side edges **132a**, **132b** of the fan portions **132** to the first end **134** of the fan portions **132** collinear with the side edges **132a**, **132b**. It should be understood that dividing lines D are shown in FIG. **1** for clarity and

that the fan portions **132** are integral with the bridge portions **140**. The first end **134** of the fan portion **132**, which extends between adjacent bridge portions **140**, defines an inner length of the fan portion **132**. Due to the geometry of the fan portion **132** tapering toward the center axis C between the first side edge **132a** and the second side edge **132b**, the second end **136** of the fan portion **132** defines an outer length of the fan portion **132** that is greater than the inner length of the fan portion **132**.

(25) Moreover, each fan portion **132** has a pair of corners **132c** defined by an intersection of the second end **136** and each of the first side edge **132a** and the second side edge **132b** of the fan portion **132**. In embodiments, the corners **132c** are formed at an angle equal to or less than 90 degrees. In other embodiments, the corners **132c** are formed at an acute angle.

(26) As shown in FIG. 1, each fan portion **132** has a first side length defined by a distance between the first end **134** of the fan portion **132** and the second end **136** of the fan portion **132** along the first side edge **132a** and the dividing line D that is collinear with the first side edge **132a**. Each fan portion **132** also has a second side length defined by a distance between the first end **134** of the fan portion **132** and the second end **136** of the fan portion **132** along the second side edge **132b** and the dividing line D that is collinear with the second side edge **132b**. In embodiments, the first side length is greater than the second side length of the fan portion **132** such that the first electrode **106** has an ellipsoid geometry.

(27) The second end **136**, the first side edge **132a** and the second side edge **132b** of each fan portion **132**, and the bridge portions **140** interconnecting the fan portions **132** define an outer perimeter **138** of the first electrode **106**. In embodiments, a central opening **146** is formed within the first electrode **106** between the fan portions **132** and the bridge portions **140**, and is coaxial with the center axis C. Each fan portion **132** has a fan length extending from a perimeter **142** of the central opening **146** to the second end **136** of the fan portion **132**. Each bridge portion **140** has a bridge length extending from a perimeter **142** of the central opening **146** to the end **140a** of the bridge portion **140**, i.e., the channel **133**. As shown, the bridge length of each of the bridge portions **140** is substantially equal to one another. Each channel **133** has a channel length defined by a distance between the end **140a** of the bridge portion **140** and the second end of the fan portion **132**. Due to the bridge length of each of the bridge portions **140** being substantially equal to one another and the first side length of the fan portions **132** being greater than the second side length of the fan portions **132**, a first pair of opposite channels **133** has a channel length greater than a channel length of a second pair of opposite channels **133**. As shown, a width of the channel **133** extending between opposing side edges **132a**, **132b** of adjacent fan portions **132** remains substantially constant due to opposing side edges **132a**, **132b** being substantially parallel to one another.

(28) In embodiments, the central opening **146** has a radius of 2 centimeters (cm) to 5 cm. In embodiments, the central opening **146** has a radius of 3 cm to 4 cm. In embodiments, a total fan area of each of the fan portions **132** is equal to or greater than twice an area of the central opening **146**. It should be appreciated that the ratio between the total fan area of the fan portions **132** and the area of the central opening **146** is directly related to a total amount of deflection of the first film layer **122** when the artificial muscle **100** is actuated, as discussed herein. In embodiments, the bridge length is 20% to 50% of the fan length. In embodiments, the bridge length is 30% to 40% of the fan length. In embodiments in which the first electrode **106** does not include the central opening **146**, the fan length and the bridge length may be measured from a perimeter of an imaginary circle coaxial with the center axis C.

(29) Similar to the first electrode **106**, the second electrode **108** includes two or more fan portions **154** extending radially from the center axis C of the artificial muscle **100**. The second electrode **108** includes substantially the same structure as the first electrode **106** and, thus, includes the same number of fan portions **154**. Specifically, the second electrode **108** is illustrated as including four fan portions **154**. However, it should be appreciated that the second electrode **108** may include any suitable number of fan portions **154**.

(30) Each fan portion **154** of the second electrode **108** has a first side edge **154a** and an opposite second side edge **154b**. As shown, the second terminal **152** extends from the second end **158** of one of the fan portions **154** and is integrally formed therewith. A channel **155** is at least partially defined by opposing side edges **154a**, **154b** of adjacent fan portions **154** and, thus, extends radially toward the center axis C. The channel **155** terminates at an end **162a** of a bridge portion **162** interconnecting adjacent fan portions **154**.

(31) As shown in FIG. 1, additional dividing lines D are included to depict the boundary between the fan portions **154** and the bridge portions **162**. The dividing lines D extend from the side edges **154a**, **154b** of the fan portions **154** to the first end **156** of the fan portions **154** collinear with the side edges **154a**, **154b**. It should be understood that dividing lines D are shown in FIG. 1 for clarity and that the fan portions **154** are integral with the bridge portions **162**. The first end **156** of the fan portion **154**, which extends between adjacent bridge portions **162**, defines an inner length of the fan portion **154**. Due to the geometry of the fan portion **154** tapering toward the center axis C between the first side edge **154a** and the second side edge **154b**, the second end **158** of the fan portion **154** defines an outer length of the fan portion **154** that is greater than the inner length of the fan portion **154**.

(32) Moreover, each fan portion **154** has a pair of corners **154c** defined by an intersection of the second end **158** and each of the first side edge **154a** and the second side edge **154b** of the fan portion **154**. In embodiments, the corners **154c** are formed at an angle equal to or less than 90 degrees. In other embodiments, the corners **154c** are formed at an acute angle. As described in more detail herein, during actuation of the artificial muscle **100**, the corners **132c** of the first electrode **106** and the corners **154c** of the second electrode **108** are configured to be attracted to one another at a lower voltage as compared to the rest of the first electrode **106** and the second electrode **108**. Thus, actuation of the artificial muscle **100** initially at the corners **132c**, **154c** results the outer perimeter **138** of the first electrode **106** and the outer perimeter **160** of the second electrode **108** being attracted to one another at a lower voltage and reducing the likelihood of air pockets or voids forming between the first electrode **106** and the second electrode **108** after actuation of the artificial muscle **100**.

(33) As shown in FIGS. 1 and 2, in embodiments, the first side edge **154a** of each fan portion **154** has a first side length defined by a distance between the first end **156** of the fan portion **154** and the second end **158** of the fan portion **154** along the first side edge **154a** and the dividing line D that is collinear with the first side edge **154a**. Each fan portion **154** also has a second side length defined by a distance between the first end **156** of the fan portion **154** and the second end **158** of the fan portion **154** along the second side edge **154b** and the dividing line D that is collinear with the second side edge **154b**. In embodiments, the first side length is greater than the second side length of the fan portion **154** such that the second electrode **108** has an ellipsoid geometry corresponding to the geometry of the first electrode **106**.

(34) The second end **158**, the first side edge **154a** and the second side edge **154b** of each fan portion **154**, and the bridge portions **162** interconnecting the fan portions **154** define an outer perimeter **160** of the second electrode **108**. In embodiments, a central opening **168** is formed within the second electrode **108** between the fan portions **154** and the bridge portions **162**, and is coaxial with the center axis C. Each fan portion **154** has a fan length extending from a perimeter **164** of the central opening **168** to the second end **158** of the fan portion **154**. Each bridge portion **162** has a bridge length extending from the central opening **168** to the end **162a** of the bridge portion **162**, i.e., the channel **155**. As shown, the bridge length of each of the bridge portions **162** is substantially equal to one another. Each channel **155** has a channel length defined by a distance between the end **162a** of the bridge portion **162** and the second end of the fan portion **154**. Due to the bridge length of each of the bridge portions **162** being substantially equal to one another and the first side length of the fan portions **154** being greater than the second side length of the fan portions **154**, a first pair of opposite channels **155** has a channel length greater than a channel length of a second pair of

opposite channels **155**. As shown, a width of the channel **155** extending between opposing side edges **154a**, **154b** of adjacent fan portions **154** remains substantially constant due to opposing side edges **154a**, **154b** being substantially parallel to one another.

(35) In embodiments, the central opening **168** has a radius of 2 cm to 5 cm. In embodiments, the central opening **168** has a radius of 3 cm to 4 cm. In embodiments, a total fan area of each of the fan portions **154** is equal to or greater than twice an area of the central opening **168**. It should be appreciated that the ratio between the total fan area of the fan portions **154** and the area of the central opening **168** is directly related to a total amount of deflection of the second film layer **124** when the artificial muscle **100** is actuated. In embodiments, the bridge length is 20% to 50% of the fan length. In embodiments, the bridge length is 30% to 40% of the fan length. In embodiments in which the second electrode **108** does not include the central opening **168**, the fan length and the bridge length may be measured from a perimeter of an imaginary circle coaxial with the center axis C.

(36) As described herein, the first electrode **106** and the second electrode **108** each have a central opening **146**, **168** coaxial with the center axis C. However, it should be understood that the first electrode **106** does not need to include the central opening **146** when the central opening **168** is provided within the second electrode **108**. Alternatively, the second electrode **108** does not need to include the central opening **168** when the central opening **146** is provided within the first electrode **106**.

(37) Referring again to FIG. 1, the electrical insulator membrane **110** may have a substantially ellipsoid geometry generally corresponding to the geometry of the first electrode **106** and the second electrode **108**. Thus, the electrical insulator membrane **110** may have fan portions **172** and bridge portions **176** corresponding to like portions on the first electrode **106** and the second electrode **108**. Further, the electrical insulator membrane **110** has an outer perimeter **180** corresponding to the outer perimeter **138** of the first electrode **106** and the outer perimeter **160** of the second electrode **108**, respectively, when positioned thereon. In embodiments, the electrical insulator membrane **110** may extend across the central opening **146**, **168** of the first electrode **106** and the second electrode **108**. In other embodiments, the electrical insulator membrane **110** may also have a central opening formed therein and coaxial with the central opening **146**, **168** of the first electrode **106** and the second electrode **108**. Additionally, the electrical insulator membrane **110** may completely isolate the second electrode **108**, or in embodiments, the first electrode **106**, from a dielectric fluid **198**.

(38) In embodiments, the electrical insulator membrane **110** includes an adhesive surface **184** and an opposite non-sealable surface **188**. Thus, in embodiments, the electrical insulator membrane **110** is a polymer tape adhered to the inner surface **150** of the second electrode **108**. In embodiments, the electrical insulator membrane **110** comprises BOPP.

(39) Referring now to FIGS. 2-4, the artificial muscle **100** is shown in its assembled form with the first terminal **130** of the first electrode **106** and the second terminal **152** of the second electrode **108** extending past an outer perimeter of the housing **102**, i.e., the first film layer **122** and the second film layer **124**. As shown in FIG. 2, the second electrode **108** is stacked on top of the first electrode **106** and, therefore, the first electrode **106**, the first film layer **122**, and the second film layer **124** are not shown. In its assembled form, the first electrode **106**, the second electrode **108**, and the electrical insulator membrane **110** are sandwiched between the first film layer **122** and the second film layer **124**. The first film layer **122** is partially sealed to the second film layer **124** at an area surrounding the outer perimeter **138** of the first electrode **106** and the outer perimeter **160** of the second electrode **108**. In some embodiments, the first film layer **122** is heat-sealed to the second film layer **124**. Specifically, in some embodiments, the first film layer **122** is sealed to the second film layer **124** to define a sealed portion **190** surrounding the first electrode **106** and the second electrode **108**. The first film layer **122** and the second film layer **124** may be sealed in any suitable manner, such as using an adhesive, heat sealing, vacuum sealing, or the like.

(40) The first electrode **106**, the second electrode **108**, and the electrical insulator membrane **110** provide a barrier that prevents the first film layer **122** from sealing to the second film layer **124** forming an unsealed portion **192**. The unsealed portion **192** of the housing **102** includes an electrode region **194**, in which the electrode pair **104** is provided, and an expandable fluid region **196**, which is surrounded by the electrode region **194**. The central openings **146**, **168** of the first electrode **106** and the second electrode **108** define the expandable fluid region **196** and are arranged to be axially stacked on one another. Although not shown, the housing **102** may be cut to conform to the geometry of the electrode pair **104** and reduce the size of the artificial muscle **100**, namely, the size of the sealed portion **190**.

(41) A dielectric fluid **198** is provided within the unsealed portion **192** and flows freely between the first electrode **106** and the second electrode **108**. A “dielectric” fluid as used herein is a medium or material that transmits electrical force without conduction and as such has low electrical conductivity. Some non-limiting example dielectric fluids include perfluoroalkanes, transformer oils, and deionized water. It should be appreciated that the dielectric fluid **198** may be injected into the unsealed portion **192** of the artificial muscle **100** using a needle or other suitable injection device.

(42) Referring now to FIGS. **3** and **4**, the artificial muscle **100** is actuatable between a non-actuated state and an actuated state. In the non-actuated state, as shown in FIG. **3**, the first electrode **106** and the second electrode **108** are partially spaced apart from one another proximate the central openings **146**, **168** thereof and the first end **134**, **156** of the fan portions **132**, **154**. The second end **136**, **158** of the fan portions **132**, **154** remain in position relative to one another due to the housing **102** being sealed at the outer perimeter **138** of the first electrode **106** and the outer perimeter **160** of the second electrode **108**. In the actuated state, as shown in FIG. **4**, the first electrode **106** and the second electrode **108** are brought into contact with and oriented parallel to one another to force the dielectric fluid **198** into the expandable fluid region **196**. This causes the dielectric fluid **198** to flow through the central openings **146**, **168** of the first electrode **106** and the second electrode **108** and inflate the expandable fluid region **196**.

(43) Referring now to FIG. **3**, the artificial muscle **100** is shown in the non-actuated state. The electrode pair **104** is provided within the electrode region **194** of the unsealed portion **192** of the housing **102**. The central opening **146** of the first electrode **106** and the central opening **168** of the second electrode **108** are coaxially aligned within the expandable fluid region **196**. As such, the central openings **146**, **168** encircle and define the expandable fluid region **196**. In the non-actuated state, the first electrode **106** and the second electrode **108** are partially spaced apart from and non-parallel to one another. Due to the first film layer **122** being sealed to the second film layer **124** around the electrode pair **104**, the second end **136**, **158** of the fan portions **132**, **154** are brought into contact with one another. Thus, dielectric fluid **198** is provided between the first electrode **106** and the second electrode **108**, thereby separating the first end **134**, **156** of the fan portions **132**, **154** proximate the expandable fluid region **196**. Stated another way, a distance between the first end **134** of the fan portion **132** of the first electrode **106** and the first end **156** of the fan portion **154** of the second electrode **108** is greater than a distance between the second end **136** of the fan portion **132** of the first electrode **106** and the second end **158** of the fan portion **154** of the second electrode **108**. This results in the electrode pair **104** zippering toward the expandable fluid region **196** when actuated. More particularly, zippering of the electrode pair **104** is initiated at the corners **132c** of the first electrode **106** and the corners **154c** of the second electrode **108**, as discussed herein. In some embodiments, the first electrode **106** and the second electrode **108** may be flexible. Thus, as shown in FIG. **3**, the first electrode **106** and the second electrode **108** are convex such that the second ends **136**, **158** of the fan portions **132**, **154** thereof may remain close to one another, but spaced apart from one another proximate the central openings **146**, **168**. In the non-actuated state, the expandable fluid region **196** has a first height **H1**.

(44) When actuated, as shown in FIG. **4**, the first electrode **106** and the second electrode **108** zipper

toward one another from the second ends **136**, **158** of the fan portions **132**, **154** thereof, thereby pushing the dielectric fluid **198** into the expandable fluid region **196**. As shown, when in the actuated state, the first electrode **106** and the second electrode **108** are parallel to one another. In the actuated state, the dielectric fluid **198** flows into the expandable fluid region **196** to inflate the expandable fluid region **196**. As such, the first film layer **122** and the second film layer **124** expand in opposite directions. In the actuated state, the expandable fluid region **196** has a second height H_2 , which is greater than the first height H_1 of the expandable fluid region **196** when in the non-actuated state. Although not shown, it should be noted that the electrode pair **104** may be partially actuated to a position between the non-actuated state and the actuated state. This would allow for partial inflation of the expandable fluid region **196** and adjustments when necessary.

(45) In order to move the first electrode **106** and the second electrode **108** toward one another, a voltage is applied by a power supply, such as the power supply **406** (FIG. 9). In some embodiments, a voltage of up to 10 kV may be provided from the power supply to induce an electric field through the dielectric fluid **198**. The resulting attraction between the first electrode **106** and the second electrode **108** pushes the dielectric fluid **198** into the expandable fluid region **196**. Pressure from the dielectric fluid **198** within the expandable fluid region **196** causes the first film layer **122** to deform in a first axial direction along the center axis C of the first electrode **106** and causes the second film layer **124** and the electrical insulator membrane **110** to deform in an opposite second axial direction along the center axis C of the second electrode **108**. Once the voltage being supplied to the first electrode **106** and the second electrode **108** is discontinued, the first electrode **106** and the second electrode **108** return to their initial, non-parallel position in the non-actuated state.

(46) It should be appreciated that the present embodiments disclosed herein, specifically, the fan portions **132**, **154** with the interconnecting bridge portions **140**, **162**, provide a number of improvements over actuators, such as HASEL actuators, that do not include the fan portions **132**, **154**. Embodiments of the artificial muscle **100** including fan portions **132**, **154** on each of the first electrode **106** and the second electrode **108**, respectively, increases the surface area and, thus, displacement at the expandable fluid region **196** without increasing the amount of voltage required as compared to known HASEL actuators including donut-shaped electrodes having a uniform, radially-extending width. In addition, the corners **132c**, **154c** of the fan portions **132**, **154** of the artificial muscle **100** provide zipping fronts that result in focused and directed zipping along the outer perimeters **138**, **160** of the first electrode **106** and the second electrode **108** during actuation as compared to HASEL actuators including donut-shaped electrodes.

(47) Specifically, one pair of fan portions **132**, **154** provides at least twice the amount of actuator power per unit volume as compared to donut-shaped HASEL actuators, while two pairs of fan portions **132**, **154** provide at least four times the amount of actuator power per unit volume. The bridge portions **140**, **162** interconnecting the fan portions **132**, **154** also limit buckling of the fan portions **132**, **154** by maintaining the distance between the channels **133**, **155** and the central openings **146**, **168**. Because the bridge portions **140**, **162** are integrally formed with the fan portions **132**, **154**, the bridge portions **140**, **162** also prevent tearing and leakage between the fan portions **132**, **154** by eliminating attachment locations that provide an increased risk of rupturing.

(48) In operation, when the artificial muscle **100** is actuated, expansion of the expandable fluid region **196** produces a force of 20 Newton-millimeters (N.Math.mm) per cubic centimeter (cm.sup.3) of actuator volume or greater, such as 25 N.Math.mm per cm.sup.3 or greater, 30 N.Math.mm per cm.sup.3 or greater, 35 N.Math.mm per cm.sup.3 or greater, 40 N.Math.mm per cm.sup.3 or greater, or the like. In one example, when the artificial muscle **100** is actuated by a voltage of 9.5 kilovolts (kV), the artificial muscle **100** provides a resulting force of 20 N.

(49) Moreover, the size of the first electrode **106** and the second electrode **108** is proportional to the amount of displacement of the dielectric fluid **198**. Therefore, when greater displacement within the expandable fluid region **196** is desired, the size of the electrode pair **104** is increased relative to

the size of the expandable fluid region **196**. It should be appreciated that the size of the expandable fluid region **196** is defined by the central openings **146**, **168** in the first electrode **106** and the second electrode **108**. Thus, the degree of displacement within the expandable fluid region **196** may alternatively, or in addition, be controlled by increasing or reducing the size of the central openings **146**, **168**.

(50) Referring now to FIG. 5, a cross-sectional view of the artificial muscle **100** is depicted taken along line 5-5 in FIG. 3. As shown in FIG. 5, the electrical insulator membrane **110** is depicted in greater detail illustrating a plurality of individual layers. More particularly, in embodiments, the electrical insulator membrane **110** includes a pair of electrical insulator layers, such as a first electrical insulator layer **111** and a second electrical insulator layer **112**, a first adhesive layer **113** provided between the first electrical insulator layer **111** and the second electrical insulator layer **112**, and a second adhesive layer **115** provided between the second electrical insulator layer **112** and the second electrode **108**. The first electrical insulator layer **111** and the second electrical insulator layer **112** may be formed from BOPP. The first adhesive layer **113** and the second adhesive layer **115** may be formed from any suitable material such as, for example, acrylic or the like.

(51) It should be appreciated that repeated use of an artificial muscle including a non-insulated electrode electrically coupled to a positive terminal of a power supply will result in oscillating force output and a decreasing force output over time. Accordingly, the positive terminal **406A** of the power supply **406** it so be electrically coupled to the insulated electrode, i.e., the second electrode **108**.

(52) Referring now to FIG. 6, a partial cross-sectional view of an artificial muscle **200** is illustrated. It should be appreciated that the artificial muscle **200** is similar to the artificial muscle **100** and, thus, like reference numerals will be used to refer to like parts. As with the artificial muscle **100**, the second electrode **108** is coated by an electrical insulator membrane **210**, thereby leaving the first electrode **106** non-insulated. In embodiments, the electrical insulator membrane **210** includes a pair of electrical insulator layers, such as the first electrical insulator layer **111** and the second electrical insulator layer **112**, the first adhesive layer **113**, and the second adhesive layer **115**. The first adhesive layer **113** and the second adhesive layer **115** are both provided between the first electrical insulator layer **111** and the second electrical insulator layer **112**. Accordingly, the first adhesive layer **113** and the second adhesive layer **115** are in contact with one another and sandwiched between the first electrical insulator layer **111** and the second electrical insulator layer **112**, thereby increasing a distance between the first electrical insulator layer **111** and the second electrical insulator layer **112**.

(53) It should be appreciated that the electrical insulator membranes **110**, **210** illustrated in FIGS. 5 and 6, respectively, may include only a single electrical insulator layer as opposed to the pair of electrical insulator layers **111**, **112**. However, the pair of electrical insulator layers **111**, **112** provides improved integrity and reduced likelihood of failure when one of the electrical insulator layers **111**, **112** develops a defect.

(54) Furthermore, in embodiments, it should be appreciated that each of the first electrode **106** and the second electrode **108** may include a plurality of electrode segments as opposed to a singular, integrally formed electrode extending along the entire periphery of the artificial muscle **100**. In this embodiment, one or more segments of the first electrode **106** and the second electrode **108** may be insulated if one or more corresponding, opposite electrode segments of the other of the first electrode **106** and the second electrode **108** are non-insulated. However, the one or more insulated electrode segment are to be electrically coupled to the positive terminal **406A** of the power supply **406** (FIG. 9) and the one or more non-insulated electrode segments are to be electrically coupled to the negative terminal **406B** of the power supply **406** (FIG. 9). Additionally, it should be appreciated that any non-insulated electrode segment of one of the electrodes **106**, **108** should be spaced apart from any non-insulated electrode segment of the other one of the electrodes **106**, **108**. Otherwise,

risk of electrical arcing and failure may occur.

(55) With more particularity and referring now to FIG. 7, a partial plan view of an artificial muscle **300** is shown with certain components hidden from view. It should be appreciated that the artificial muscle **300** is similar to the artificial muscle **100** depicted herein and, thus, like reference numerals will be used to refer to like parts. Rather than the artificial muscle **300** including the first electrode **106** and the second electrode **108**, each of which being a one-piece integrally formed electrode, the artificial muscle **300** includes a first electrode **306** (FIG. 8) and a second electrode **308**, each including a plurality of electrode segments. For example, as shown in FIG. 7, the second electrode **308** includes a plurality of electrode segments **308a**, **308b**, **308c**, **308d**, **308e**, **308f**. Additionally, each of the electrode segments **308a**, **308b**, **308c**, **308d**, **308e**, **308f** includes a terminal **130** extending from an outer edge thereof and through the housing **102**.

(56) Referring still to FIG. 7, electrode segments **308a**, **308c**, **308e** are provided with an electrical insulator membrane **310**, such as the electrical insulator membrane **110**, and the remaining electrode segments **308b**, **308d**, **308f** are left exposed, i.e., non-insulated. It should be appreciated that the electrode segments **308a**, **308b**, **308c**, **308d**, **308e**, **308f** alternate between those that are electrically insulated by the electrical insulator membrane **310** and those that are not electrically insulated by the electrical insulator membrane **310**. As such, the electrical insulator membrane **310** for each electrode segment **308a**, **308c**, **308e** is separate from one another to avoid insulating adjacent electrode segments. It should be appreciated that the electrode segments **308a**, **308c**, **308e** that are electrically insulated by the electrical insulator membrane **310** are electrically coupled to the positive terminal **406A** of the power supply **406** (FIG. 9). Additionally, the electrode segments **308b**, **308d**, **308f** that are not electrically insulated by the electrical insulator membrane **310** are electrically coupled to the negative terminal **406B** of the power supply **406** (FIG. 9).

(57) Referring now to FIG. 8, a cross-sectional view of the artificial muscle **300** is illustrated. Although only a pair of electrode segments **306a**, **306d** of the first electrode **306** are illustrated, it should be appreciated that the first electrode **306** may be a mirror image of the second electrode **308** when positioned such that the electrode segments **306a**, **306d** of the first electrode **306** overlap a corresponding one of the electrode segments **308a**, **308d** of the second electrode **308** defining overlapping pairs of electrode segments. However, only one electrode segment of each pair of electrode segments is electrically insulated by the electrical insulator membrane **310** while the opposite electrode segment is not electrically insulated by the electrical insulator membrane **310**. For example, electrode segment **308a** is insulated by the electrical insulator membrane **310** while electrode segment **306a** is not insulated by the electrical insulator membrane **310**. Additionally, electrode segment **308d** not insulated by the electrical insulator membrane **310** while electrode segment **306d** is insulated by the electrical insulator membrane **310**. Similar to the second electrode **308**, it should be appreciated that the electrode segments **306d** that is electrically insulated by the electrical insulator membrane **310** is electrically coupled to the positive terminal **406A** of the power supply **406** (FIG. 9) and the electrode segment **306a** that is not electrically insulated by the electrical insulator membrane **310** is electrically coupled to the negative terminal **406B** of the power supply **406** (FIG. 9).

(58) Referring now to FIG. 9, an actuation system **400** may be provided for operating an artificial muscle, such as the artificial muscles **100**, **200**, **300**, between the non-actuated state and the actuated state. Thus, the actuation system **400** may include a controller **402**, an operating device **404**, a power supply **406**, and a communication path **408**. The various components of the actuation system **400** will now be described.

(59) The controller **402** includes a processor **410** and a non-transitory electronic memory **412** to which various components are communicatively coupled. In some embodiments, the processor **410** and the non-transitory electronic memory **412** and/or the other components are included within a single device. In other embodiments, the processor **410** and the non-transitory electronic memory **412** and/or the other components may be distributed among multiple devices that are

communicatively coupled. The controller **402** includes non-transitory electronic memory **412** that stores a set of machine-readable instructions. The processor **410** executes the machine-readable instructions stored in the non-transitory electronic memory **412**. The non-transitory electronic memory **412** may comprise RAM, ROM, flash memories, hard drives, or any device capable of storing machine-readable instructions such that the machine-readable instructions can be accessed by the processor **410**. Accordingly, the actuation system **400** described herein may be implemented in any conventional computer programming language, as pre-programmed hardware elements, or as a combination of hardware and software components. The non-transitory electronic memory **412** may be implemented as one memory module or a plurality of memory modules.

(60) In some embodiments, the non-transitory electronic memory **412** includes instructions for executing the functions of the actuation system **400**. The instructions may include instructions for operating the artificial muscles **100, 200, 300** based on a user command.

(61) The processor **410** may be any device capable of executing machine-readable instructions. For example, the processor **410** may be an integrated circuit, a microchip, a computer, or any other computing device. The non-transitory electronic memory **412** and the processor **410** are coupled to the communication path **408** that provides signal interconnectivity between various components and/or modules of the actuation system **400**. Accordingly, the communication path **408** may communicatively couple any number of processors with one another, and allow the modules coupled to the communication path **408** to operate in a distributed computing environment. Specifically, each of the modules may operate as a node that may send and/or receive data. As used herein, the term “communicatively coupled” means that coupled components are capable of exchanging data signals with one another such as, for example, electrical signals via conductive medium, electromagnetic signals via air, optical signals via optical waveguides, and the like.

(62) As schematically depicted in FIG. 9, the communication path **408** communicatively couples the processor **410** and the non-transitory electronic memory **412** of the controller **402** with a plurality of other components of the actuation system **400**. For example, the actuation system **400** depicted in FIG. 9 includes the processor **410** and the non-transitory electronic memory **412** communicatively coupled with the operating device **404** and the power supply **406**.

(63) The operating device **404** allows for a user to control operation of the artificial muscles **100, 200, 300**. In some embodiments, the operating device **404** may be a switch, toggle, button, or any combination of controls to provide user operation. As a non-limiting example, a user may actuate the artificial muscles **100, 200, 300** into the actuated state by activating controls of the operating device **404** to a first position. While in the first position, the artificial muscles **100, 200, 300** will remain in the actuated state. The user may switch the artificial muscles **100, 200, 300** into the non-actuated state by operating the controls of the operating device **404** out of the first position and into a second position.

(64) The operating device **404** is coupled to the communication path **408** such that the communication path **408** communicatively couples the operating device **404** to other modules of the actuation system **400**. The operating device **404** may provide a user interface for receiving user instructions as to a specific operating configuration of the artificial muscles **100, 200, 300**. In addition, user instructions may include instructions to operate the artificial muscles **100, 200, 300** only at certain conditions.

(65) The power supply **406** (e.g., battery) provides power to the artificial muscles **100, 200, 300**. In some embodiments, the power supply **406** is a rechargeable direct current power source. It is to be understood that the power supply **406** may be a single power supply or battery for providing power to the artificial muscle **100, 200, 300**. A power adapter (not shown) may be provided and electrically coupled via a wiring harness or the like for providing power to the artificial muscles **100, 200, 300** via the power supply **406**. As discussed herein, the power supply **406** includes a positive terminal **406A** and a negative terminal **406B**. The positive terminal **406A** is electrically coupled to the insulated electrode, or the insulated electrode segments, of the first electrode **106**

and the second electrode **108** and the negative terminal **406B** is electrically coupled to the non-insulated electrode, or non-insulated electrode segments, of the first electrode **106** and the second electrode **108**. In the embodiments discussed herein, the positive terminal **406A** is electrically coupled to the second terminal **152** of the second electrode **108** and the negative terminal **406B** is electrically coupled to the first terminal **130** of the first electrode **106**.

(66) In some embodiments, the actuation system **400** also includes a display device **414**. The display device **414** is coupled to the communication path **408** such that the communication path **408** communicatively couples the display device **414** to other modules of the actuation system **400**. The display device **414** may output a notification in response to an actuation state of the artificial muscles **100**, **200**, **300** or indication of a change in the actuation state of the artificial muscles **100**, **200**, **300**. Moreover, the display device **414** may be a touchscreen that, in addition to providing optical information, detects the presence and location of a tactile input upon a surface of or adjacent to the display device **414**. Accordingly, the display device **414** may include the operating device **404** and receive mechanical input directly upon the optical output provided by the display device **414**.

(67) In some embodiments, the actuation system **400** includes network interface hardware **416** for communicatively coupling the actuation system **400** to a portable device **418** via a network **420**. The portable device **418** may include, without limitation, a smartphone, a tablet, a personal media player, or any other electric device that includes wireless communication functionality. It is to be appreciated that, when provided, the portable device **418** may serve to provide user commands to the controller **402**, instead of the operating device **404**. As such, a user may be able to control or set a program for controlling the artificial muscles **100**, **200**, **300** without utilizing the controls of the operating device **404**. Thus, the artificial muscles **100**, **200**, **300** may be controlled remotely via the portable device **418** wirelessly communicating with the controller **402** via the network **420**.

(68) From the above, it is to be appreciated that defined herein are artificial muscles for inflating or deforming a surface of an object by selectively actuating the artificial muscle to raise and lower a region thereof. This provides a low profile inflation member that may operate on demand.

(69) It is noted that the terms “substantially” and “about” may be utilized herein to represent the inherent degree of uncertainty that may be attributed to any quantitative comparison, value, measurement, or other representation. These terms are also utilized herein to represent the degree by which a quantitative representation may vary from a stated reference without resulting in a change in the basic function of the subject matter at issue.

(70) While particular embodiments have been illustrated and described herein, it should be understood that various other changes and modifications may be made without departing from the scope of the claimed subject matter. Moreover, although various aspects of the claimed subject matter have been described herein, such aspects need not be utilized in combination. It is therefore intended that the appended claims cover all such changes and modifications that are within the scope of the claimed subject matter.

Claims

1. An artificial muscle system comprising: an artificial muscle comprising: an electrode pair comprising a first electrode and a second electrode; an electrical insulator membrane fixed to a surface of the second electrode facing the first electrode, the electrical insulator membrane comprising a first electrical insulator layer and a second electrical insulator layer positioned between the first electrical insulator layer and the second electrode; a housing comprising an electrode region and an expandable fluid region; the electrode pair positioned in the electrode region of the housing; and a dielectric fluid housed within the housing; and a power supply including a positive terminal and a negative terminal, the positive terminal electrically coupled to the second electrode, the negative terminal electrically coupled to the first electrode, wherein the

electrode pair is actuatable between a non-actuated state and an actuated state such that actuation from the non-actuated state to the actuated state directs the dielectric fluid into the expandable fluid region.

2. The artificial muscle system of claim 1, wherein the electrical insulator membrane comprises: a first adhesive layer positioned between the first electrical insulator layer and the second electrical insulator layer.

3. The artificial muscle system of claim 2, wherein the electrical insulator membrane comprises: a second adhesive layer positioned between the first electrical insulator layer and the second electrical insulator layer.

4. The artificial muscle system of claim 2, wherein the electrical insulator membrane comprises: a second adhesive layer positioned between the second electrical insulator layer and the second electrode.

5. The artificial muscle system of claim 1, wherein the first electrical insulator layer and the second electrical insulator layer each comprises biaxially oriented polypropylene.

6. The artificial muscle system of claim 1, wherein a surface of the first electrode is in direct contact with the dielectric fluid.

7. The artificial muscle system of claim 1, wherein the first electrode and the second electrode each comprise two or more fan portions and two or more bridge portions, wherein: each of the two or more bridge portions interconnects adjacent fan portions; and at least one of the first electrode and the second electrode comprises a central opening positioned between the two or more fan portions and encircling the expandable fluid region.

8. An artificial muscle system comprising: an artificial muscle comprising: an electrode pair comprising a first electrode and a second electrode, the first electrode and the second electrode each comprising a plurality of electrode segments; an electrical insulator membrane fixed to one or more insulated electrode segments of the first electrode and one or more opposite insulated electrode segments of the second electrode; a housing comprising an electrode region and an expandable fluid region; the electrode pair positioned in the electrode region of the housing, the first electrode fixed to a first surface of the housing and the second electrode fixed to a second surface of the housing; and a dielectric fluid housed within the housing; and a power supply including a positive terminal and a negative terminal, the positive terminal electrically coupled to the one or more insulated electrode segments of the first electrode and the one or more opposite insulated electrode segments of the second electrode, the negative terminal electrically coupled to one or more non-insulated electrode segments of the first electrode and one or more opposite non-insulated electrode segments of the second electrode, wherein the electrode pair is actuatable between a non-actuated state and an actuated state such that actuation from the non-actuated state to the actuated state directs the dielectric fluid into the expandable fluid region.

9. The artificial muscle system of claim 8, wherein the electrical insulator membrane is not fixed to the one or more non-insulated electrode segments of the first electrode and the one or more non-insulated electrode segments of the second electrode.

10. The artificial muscle system of claim 8, wherein the one or more insulated electrode segments of the first electrode and the one or more insulated electrode segments of the second electrode are spaced apart from one another by the one or more non-insulated electrode segments of the first electrode and the one or more non-insulated electrode segments of the second electrode, respectively.

11. The artificial muscle system of claim 9, wherein the electrical insulator membrane comprises: a first electrical insulator layer; and a second electrical insulator layer positioned between the first electrical insulator layer and the second electrode.

12. The artificial muscle system of claim 11, wherein the electrical insulator membrane comprises: a first adhesive layer positioned between the first electrical insulator layer and the second electrical insulator layer.

13. The artificial muscle system of claim 12, wherein the electrical insulator membrane comprises: a second adhesive layer positioned between the first electrical insulator layer and the second electrical insulator layer.
14. The artificial muscle system of claim 12, wherein the electrical insulator membrane comprises: a second adhesive layer positioned between the second electrical insulator layer and the second electrode.
15. The artificial muscle system of claim 11, wherein the first electrical insulator layer and the second electrical insulator layer each comprises biaxially oriented polypropylene.
16. A method for actuating an artificial muscle, the method comprising: electrically coupling a negative terminal and a positive terminal of a power supply to an electrode pair of the artificial muscle, the artificial muscle comprising: a housing having an electrode region and an expandable fluid region; the electrode pair comprising a first electrode electrically coupled to the negative terminal of the power supply, and a second electrode electrically coupled to the positive terminal of the power supply, the electrode pair positioned in the electrode region of the housing; and an electrical insulator membrane fixed to the second electrode, the electrical insulator membrane comprising a first electrical insulator layer and a second electrical insulator layer positioned between the first electrical insulator layer and the second electrode; and applying a voltage from the power supply to the electrode pair of the artificial muscle, thereby actuating the electrode pair from a non-actuated state to an actuated state such that dielectric fluid is directed into the expandable fluid region of the housing and expands the expandable fluid region.
17. The method of claim 16, wherein the electrical insulator membrane is fixed to a surface of the second electrode facing the first electrode.
18. The method of claim 17, wherein a surface of the first electrode is in direct contact with the dielectric fluid.
19. The method of claim 16, wherein the first electrode and the second electrode each comprise two or more fan portions and two or more bridge portions, wherein: each of the two or more bridge portions interconnects adjacent fan portions; and at least one of the first electrode and the second electrode comprises a central opening positioned between the two or more fan portions and encircling the expandable fluid region.
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